

# A Meta-Analysis of Shared Leadership and Team Effectiveness

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A growing number of studies have examined the “sharedness” of leadership processes in teams (i.e., shared leadership, collective leadership, and distributed leadership). We meta-analytically cumulated 42 independent samples of shared leadership and examined its relationship to team effectiveness. Our findings reveal an overall positive relationship ( $\rho = .34$ ). But perhaps more important, *what* is actually shared among members appears to matter with regard to team effectiveness. That is, shared traditional forms of leadership (e.g., initiating structure and consideration) show a lower relationship ( $\rho = .18$ ) than either shared new-genre leadership (e.g., charismatic and transformational leadership;  $\rho = .34$ ) or cumulative, overall shared leadership ( $\rho = .35$ ). In addition, shared leadership tends to be more strongly related to team attitudinal outcomes and behavioral processes and emergent team states, compared with team performance. Moreover, the effects of shared leadership are stronger when the work of team members is more complex. Our findings further suggest that the referent used in measuring shared leadership does not influence its relationship with team effectiveness and that compared with vertical leadership, shared leadership shows unique effects in relation to team performance. In total, our study not only cumulates extant research on shared leadership but also provides directions for future research to move forward in the study of plural forms of leadership.

**Keywords:** shared leadership, team effectiveness, meta-analysis

In the latter portion of the 20th century, leadership theory and research was dominated by a focus on singular, formal leaders, and how these individuals could make an impact within organizations (Bass & Bass, 2008). The specific emphasis was on understanding formal leaders in terms of behavioral qualities such as consideration, the initiation of structure, and inspiration. In more recent times, a broader perspective has emerged that depicts leadership as a process of influence occurring within organizations in which a single formal leader plays only a part (Day, 2000; Day & Harrison, 2007; Mehra, Smith, Dixon, & Robertson, 2006). Combinations of multiple *formal* leaders may need to be considered to provide a more informed picture of leadership processes (Finkelstein, Hambrick, & Cannella, 2009; Hambrick, 2007). But perhaps even more important, in addition to the leadership provided by formal individual(s) in an organization’s hierarchy (e.g., a supervisor), *informal* leadership provided by other organizational members may contribute significantly to team or unit-level effectiveness (e.g., Friedrich, Vessey, Schuelke, Ruark, & Mumford, 2009; Klein, Ziegert, Knight, & Xiao, 2006; Pearce & Sims, 2002).

For example, Day, Gronn, and Salas (2004) suggested that the overall leadership capacity of an organizational entity is a form of social capital that involves sharedness and distributed influence

among various members of the entity—formal and nonformal leaders alike. Social capital of this nature may be increasingly essential in environments in which knowledge workers are abundant, structures are flatter or more horizontal, and organizational boundaries are more open (Denis, Langley, & Sergi, 2012). Specifically, we define shared leadership in terms of an emergent team property of mutual influence and shared responsibility among team members, whereby they lead each other toward goal achievement (Carson, Tesluk, & Marrone, 2007; Hoch & Kozlowski, *in press*; Pearce & Conger, 2003). We should note that shared leadership differs from teamwork (or team processes) in that the former stresses distributed influence and responsibility among team members (Carson et al., 2007), while the latter pertains to a set of cooperatively oriented cognitions, attitudes, and actions through which team members convert member inputs to team outputs (Day et al., 2004; Guzzo & Shea, 1992; Hollenbeck, Beersma, & Schouten, 2012; LePine, Piccolo, Jackson, Mathieu, & Saul, 2008; Marks, Mathieu, & Zaccaro, 2001).

As an example of the shifting emphasis in leadership research, consider the issue of visionary leadership processes. Traditionally, visionary leadership has been conceived largely as a downward, hierarchical or vertical process (e.g., Bass, 1985). As such, the relevance of shared leadership to vision formation processes may initially appear somewhat counter to existing leadership paradigms (Contractor, DeChurch, Carson, Carter, & Keegan, 2012). However, Conger and Kanungo (1998, p. 142) purported that “the ‘visionary leader’ is more mythology than reality,” and to a large extent, the leader’s role is to capture the foresight that often exists throughout an organization. There is indeed evidence to suggest that vision formation can, and oftentimes should, be more of a process involving multiple individuals (Ensley, Hmieleski, & Pearce, 2006; Nanus, 1992; Pearce & Ensley, 2004; Yukl, 2010).

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In a similar vein, Vera and Crossan (2004, p. 227) stated that “the ideal leader might recognize his or her limitations and share the leadership of organizational learning with colleagues.” In short, more recent approaches view effective leadership as less of a top-down, leader-centric process, and more of a follower-centric process through which certain understandings of leadership (e.g., sensemaking, vision, and so forth) emerge from lower organizational levels (Uhl-Bien, Marion, & McKelvey, 2007).

Despite the work that has proceeded with shared leadership in recent times, we see the need for meta-analytic research to clarify findings to date, as well as point toward potential needs for additional research. Overall, given the potential importance of shared leadership to team effectiveness, there is a need to quantitatively review prior research findings to ascertain the criterion-related validity of shared leadership. But more specifically, we see a number of issues, such as boundary conditions (Yammarino, Salas, Serban, Shirreffs, & Shuffler, 2012), that remain to be clarified as research goes forward.

First, the effects of shared leadership may vary based on the type or content of such leadership influence processes being examined. In other words, *what* is being shared may matter to team effectiveness. Prior work has attempted to capture alternative types of shared leadership such as cumulative, overall shared leadership (e.g., Carson et al., 2007), shared visionary leadership (e.g., Pearce & Ensley, 2004), and shared authentic leadership (e.g., Hmieleski, Cole, & Baron, 2012). Although in general, we expect to find that various forms of shared leadership (i.e., in terms of content) would be associated with team effectiveness, there may be variations in the strength of these relationships. Whereas primary studies have only been able to examine one or two forms of shared leadership, our current meta-analysis affords the opportunity to compare and contrast multiple forms in their relationships to team effectiveness. In the leadership literature, various leadership styles on the part of hierarchical, singular leaders have also been shown to differ in their effects on team or follower effectiveness (Judge & Piccolo, 2004; Judge, Piccolo, & Ilies, 2004). For example, transformational leadership tends to exhibit stronger relationships with follower performance, compared with transactional leadership (Judge & Piccolo, 2004). In short, we expect different effects in terms of varying content of shared leadership. That is, we expect that the nature or content of shared leadership to some extent determines the strength of relationships with team effectiveness.

Second, we examine whether shared leadership has stronger effects on team affective outcomes and emergent states and processes, compared with team performance. Other predictors (e.g., justice climate; Roberson & Colquitt, 2005; Whitman, Caleo, Carpenter, Horner, & Bernerth, 2012) have also been shown to have differential associations with different aspects of team outcomes, such as attitudes and performance. In a similar vein, we expect that shared leadership shows different effect sizes for different types of effectiveness.

Third, we recognize that shared leadership may occur with differential effects across contexts. Various authors have pointed toward the potential importance of context, especially in terms of the actual work that is performed. Accordingly, our meta-analysis delineates contexts in terms of complex versus noncomplex work (Dust & Ziegert, 2012; Hoch & Dulebohn, 2013).

Fourth, our understanding of shared leadership and its effects may be informed by examining the referent that is used when

researchers assess shared leadership. Specifically, as described further below, prior research has used various measurement approaches that rely upon different referents, such as individual members within a team, as well as the entity as a whole. Both types of referents were originally considered by Pearce and Conger (2003). However, the criterion validity implications of using these different referents in assessing shared leadership processes is not clear in the literature. Accordingly, we explore whether using different types of referent matters in our meta-analysis.

Fifth, an important question remains unanswered pertaining to the unique contribution of shared leadership after the effect of vertical or hierarchical leadership is taken into account. It is unclear whether these two types of leadership supplement, or simply substitute for, each other. Meta-analytic findings in this regard can help not only establish the legitimacy of shared leadership research but also provide fruitful future research directions on the potential interplay between shared and vertical leadership. As such, we examine the extent to which there is an incremental contribution of shared leadership to team outcomes above and beyond vertical forms of leadership.

In sum, we address these research issues in our meta-analysis. This study contributes to the literature by providing theorists and researchers a better understanding of how to move forward with more concerted efforts to uncover the nature of shared leadership and its effects on team outcomes. Through our analyses, we begin to form conclusions and suggest further research efforts pertaining to shared leadership and its criterion validity.

## Hypotheses and Research Questions

### Shared Leadership and Team Effectiveness

The pervasive presence of self-managing (Manz & Sims, 1987) and empowered teams, as well as flatter organizational structures (Mohrman, Cohen, & Mohrman, 1995), emphasize the possible importance of shared leadership within teams. As a potentially important complement to traditional, singular forms of leadership, the key distinction between shared and traditional, vertical leadership is that the former is a “simultaneous, ongoing, mutual influential process” (Pearce, 2004, p. 48) that involves “peer, lateral, upward or downward influences of team members” (Conger & Pearce, 2003, p. 286), while the latter involves only “a downward influence on subordinates by an appointed or elected leader” (Conger & Pearce, 2003, p. 286). In addition, despite being similar to related concepts that go beyond one individual assuming the leader role, such as self-leadership (Manz & Sims, 1980), co-leadership (Rittner & Hammons, 1992), and rotated leadership (Erez, Lepine, & Elms, 2002), shared leadership emphasizes social interactions among team members (Conger & Pearce, 2003) with a collective enactment of leadership (Contractor et al., 2012; Hiller, Day, & Vance, 2006).

We expect that shared leadership, as a group-level construct and a property of the team (Carson et al., 2007), will foster positive outcomes, not only for individuals (Avolio, Jung, Murry, & Sivabramaniam, 1996) but also for teams in particular. Specifically, members with shared leadership can get empowerment and self-control either from the appointed leader or from the self-management of the team (Dumaine, 1994; Manz & Sims, 1993). Therefore, team members can become more satisfied and respon-

sible for the decision-making process (Hoch & Dulebohn, 2013). For instance, by promoting knowledge exchange among members and increasing individuals' motivation to take responsibilities, shared leadership contributes to team cohesion, team consensus, and satisfaction (Bergman et al., 2012). In addition, as an intangible resource derived from network interaction of team members, shared leadership is likely to be positively related to team performance (e.g., Boies, Lvina, & Martens, 2010; Carson et al., 2007; Pearce & Sims, 2002) by increasing team coordination and efficiency. Moreover, as noted by Day et al. (2004), shared leadership can enhance team effectiveness by increasing team social capital, including knowledge, abilities, and skills, through team information-processing and learning.

We borrow from the social identity theory of leadership (Hogg, 2001) to better understand the connection between shared leadership and team effectiveness. This theory suggests that as group membership becomes increasingly important, and members identify more strongly with the group, "effective leadership rests increasingly on the leader being considered by followers to possess prototypical properties of the group" (Hogg, van Knippenberg, & Rast, 2012, p. 264). As argued by Hogg (2001), prototypicality is not restricted to formal leaders, and instead, could apply to group or team members. Shared leadership broadens the extent of leader prototypicality since team members themselves are able to take on the leadership role. That is, members of teams with high levels of shared leadership inherently accept their own leadership role as being prototypical because it then becomes part their own social identity.<sup>1</sup> This broadened, leader prototypicality should be associated with higher levels of trust within a team, as well as members behaving in a manner that will better serve team interests (Hogg, 2001; Hogg et al., 2012).

*Hypothesis 1:* Shared leadership is positively related to team effectiveness.

### Moderators of the Shared Leadership–Team Effectiveness Relationship

There are reasons to consider content-based, contextual, and methodological moderators that may affect the magnitude of relationships between shared leadership and team effectiveness (Robbins, Ford, & Tetrick, 2012). First, as described further below, we examined content moderators that include the form of leadership that is shared in the team (e.g., shared traditional leadership, shared new-genre leadership, and cumulative, overall shared leadership), as well as the aspects of team effectiveness being considered (e.g., attitudinal outcomes, behavioral processes and emergent states, subjective performance, and objective performance). Second, we tested the complexity of work pursued by teams as a potential contextual moderator. Third, we examined different referents that researchers have used (e.g., team as the referent, each peer as the referent, and individual members rating themselves) as a potential methodological moderator. In addition, in a supplemental analysis, we consider the primary study's publication status and team type (work vs. student teams) as potential methodological moderators.

**Content of shared leadership.** Previous studies have examined various forms of shared leadership in terms of its content regarding styles or behaviors. As is the case with singular or hierarchical leadership, a team can experience alternative styles

when multiple team members assume the role of leadership (Bergman et al., 2012). For instance, some scholars have attempted to capture the distributed influence of transactional leadership (e.g., Ensley et al., 2006), transformational leadership (e.g., Boies et al., 2010), problem solving, planning and organizing, and support and consideration styles (e.g., Hiller et al., 2006). Although all of these shared forms of leadership may be associated with team effectiveness, the magnitude of relationships could vary.

In the current research, we conceptualize the content of shared leadership in three categories: (a) traditional leadership; (b) new-genre leadership; and (c) cumulative, overall shared leadership. For shared traditional leadership, we include initiating structure and consideration (Fleishman, 1953), task-oriented and participative leadership (Kahn & Katz, 1952; Katz, Maccoby, & Morse, 1950), and transactional forms of leadership. As described by Bass (1985), traditional leadership styles assume economic exchanges between leaders and members, as well as an emphasis on maintaining the status quo. For example, traditional leaders with path-goal orientations (House & Mitchell, 1974), display specific leadership behaviors (such as contingent reward) to achieve current team goals and satisfy members' needs (Bass, Waldman, Avolio, & Bebb, 1987). Attending to the task and social aspects of work, and to exchanges between leaders and followers, is essential to the effective maintenance of the status quo. Traditional leadership behavior is not inherently the sole purview of hierarchical leaders but instead, may also be shown by team members toward each other.

With that said, a new-genre form of leadership may also be shared by team members (House & Aditya, 1997). This new-genre leadership has drawn substantial researchers' attention over the past 25 years (Avolio, Walumbwa, & Weber, 2009). It emphasizes transformational, inspirational/visionary, and charismatic leadership, as well as growth-oriented (e.g., empowerment), and integrative leadership (e.g., authentic leadership), which go beyond simple, transactional exchanges between leaders and followers (Avolio et al., 2009). Moreover, at least with regard to hierarchical or vertical leaders, new-genre forms of leadership appear to be relatively universal in terms of being prototypical of leader effectiveness (House, Hanges, Javidan, Dorfman, & Gupta, 2004).

A common theme in the new-genre leadership theories is an orientation toward development and change, rather than simply maintaining the status quo. For example, both transformational leadership (Bass, 1985) and charismatic leadership (House & Shamir, 1993) describe effective leaders as those who can inspire followers to identify with new, shared visions (Judge & Piccolo, 2004). If visionary leadership is shared within a team, together members create a vision of an appealing future state and a common mental model (Pearce & Ensley, 2004). This shared visionary leadership can further enhance the members' aspirations and intrinsic motivations to pursue higher order values (Avolio et al., 2009). As another example, one can participate in decision making in the maintenance of the status quo. Indeed, empowerment includes such participation. But to be truly empowered implies that one can potentially take actions to do things differently, to be proactive, or to innovate (Kirkman & Rosen, 1999; Zhang & Bartol, 2010). We should note that similar to what is commonly

<sup>1</sup> We thank an anonymous reviewer for providing this insight.



found in the singular leadership literature (e.g., Judge & Piccolo, 2004), the shared new-genre leadership is not likely to be completely orthogonal to shared traditional leadership. However, we believe that a distinguishing feature of the former is its developmental and change orientation.

Yet another common category or form of shared leadership concerns the extent to which individuals in a team exhibit what is perceived to be cumulative, overall leadership (e.g., Carson et al., 2007). In essence, this category is assessed based on individual members' ratings through a social network approach (cf. Carson et al., 2007; Mehra et al., 2006). As described in further detail below, a team member typically provides a general, overall rating of leadership influence for each of his or her respective peers in terms of the extent to which the team relies on that person for "leadership." In research of this nature, the content of leadership is not specified for raters. Instead, there is a shared perception that, in general, members show leadership toward each other. In this case, members use their own implicit leadership theories to evaluate the extent to which others in the team exhibit leadership (Lord, Foti, & Phillips, 1982; Lord & Maher, 1991).

The above three types of shared leadership may function differently in terms of predicting team effectiveness. We expect that new-genre forms of shared leadership may be more strongly related to team effectiveness, compared with traditional forms of shared leadership. As defined above, traditional leadership depends largely on individual transactions or dyadic interactions. As such, it may not have strong cumulative (i.e., shared) effects on the team's effectiveness. Rather, traditional leadership may be more relevant to individuals' effectiveness. In contrast, the new-genre form of leadership is largely oriented toward common aspects of team functioning (e.g., shared vision) or members' autonomy and identification with the group (e.g., empowerment). Accordingly, its effects are more likely to be shown at the team level.

For example, Pearce and Ensley (2004) conceived shared visionary leadership as the extent to which team members share in the development and reinforcement of a common vision of the team's future. The process of forming common vision may require a high degree of member interaction to reach a common mental map of the future expectations of team directions or performance (Cannon-Bowers, Salas, & Converse, 1993; Nanus, 1992). This common mental map may then provide a strong basis for team effectiveness (Pearce & Ensley, 2004). Indeed, for vision to be cognitively and emotionally shared and have positive effects on team functioning, it may need to be commonly formed by multiple team members (Nanus, 1992; Vera & Crossan, 2004). Transformational leadership, which is based largely on vision and inspiration, is highly representative of the new-genre set of theories (House & Aditya, 1997). As shown by Judge and Piccolo (2004) in their meta-analysis of singular leadership, transformational leadership has a greater potential to influence team effectiveness, compared with transactional leadership. Accordingly, we anticipate similar effects when the focus is on shared leadership.

In addition, team members may simply not accept traditional forms of leadership from fellow members. For example, behaviors along the lines of transactional or task-oriented leadership may be seen as more acceptable from one's formal supervisor, as opposed to one's peers. DeRue and Ashford (2010) argued that effective leadership inherently involves follower acceptance or granting of the leader role to leaders. Perhaps traditionally oriented, shared

leadership might be resisted by team members, which could result in moderate team outcomes at best. In total, the above arguments would suggest the following:

*Hypothesis 2a:* Shared new-genre leadership has a stronger relationship with team effectiveness compared with shared traditional leadership.

A common practice in the emergent leadership literature is to assess the extent to which an individual shows leadership in a generic or overall sense (e.g., Zhang, Waldman, & Wang, 2012). Cumulative, overall shared leadership is then equivalent to the total degree of emergent leadership demonstrated by all the members in a team. It essentially involves the extent to which respective team members are viewed by others in a team as leaders in a generic sense, as opposed to reflecting some specific, leader behavior content. For example, in Carson et al. (2007), each team member was asked the extent to which his or her team relied on a respective teammate for "leadership." Scores for individuals were then cumulated to attain a shared leadership score for the whole team. Thus, in contrast to traditional and new-genre assessments of shared leadership, for studies involving cumulative, overall shared leadership, individuals do not rate leadership in terms of specific content (e.g., transformational leadership). Instead, they rate "leadership" in a more generic sense.

We expect that cumulative, overall shared leadership functions similarly, compared with shared new-genre leadership in terms of predicting team effectiveness. According to implicit leadership and cognitive categorization theories (Lord, Foti, & De Vader, 1984; Lord & Maher, 1991), individuals generate mental prototypes of effective leadership, and then use those prototypes to evaluate the behaviors and actions of others to determine whether an individual is an effective leader. If the behaviors or actions of the leader seem to be in line with the prototype, then it is likely that the leader will be perceived in a positive light. Characteristics associated with new-genre theories of leadership such as charisma and inspiration are in line with universally held prototypes of effective leadership (Dickson, Resick, & Hanges, 2006; Lord et al., 1984). Accordingly, teams that display higher cumulative, overall leadership will concomitantly demonstrate more leadership prototype characteristics that are likely to be in line with the behaviors and characteristics stressed in the new-genre theories. Thus, we expect that for cumulative perceptions of overall leadership on the part of team members, the pattern of relationships with effectiveness will be similar to that of shared, new-genre leadership.

*Hypothesis 2b:* Cumulative, overall shared leadership has a stronger relationship with team effectiveness compared with shared traditional leadership.

**Types of team effectiveness criteria.** Much of current research on shared leadership emphasizes the effectiveness of shared leadership. Both theoretical and empirical evidence suggests that shared leadership may be related to affectively based and team process outcomes, as well as performance (e.g., Bergman et al., 2012; Carson et al., 2007; Conger & Pearce, 2003). However, the strength of relationships between shared leadership and effectiveness may vary depending on the outcome of interest.

Following recent studies on team/unit-level criteria (Roberson & Colquitt, 2005; Whitman et al., 2012), we categorize team

effectiveness into four categories: (a) attitudes (e.g., job satisfaction, commitment), (b) behavioral processes and emergent states (e.g., cooperation, helping, cohesion), (c) subjective performance (e.g., subjective ratings), and (d) objective performance (e.g., productivity, actual sales). Due to the lack of studies examining team-level withdrawal (e.g., absenteeism, turnover), we did not include this category. According to the input-process-outcome model (Hackman, 1987), the relationship between shared leadership and intermediate, emergent processes (e.g., helping, trust, and cohesion) may be stronger (Day et al., 2004), compared with more distal performance outcomes. Moreover, shared leadership and affective outcomes may be mutually related and enhance each other in the team development process, which can ultimately lead to team effectiveness (Bligh, Pearce, & Kohles, 2006). Similarly, some shared leadership researchers have used team potency as a mediator in the relation to team performance (e.g., Sivasubramanian, Murry, Avolio, & Jung, 2002), because such leadership may enhance team members' confidence to succeed.

This is consistent with Yammarino et al.'s (2012) view that shared leadership, as an aspect of team process, can also be an input to other team processes or emergent properties of the team. Thus, by sharing leadership roles in a team, members work toward a common goal and lead each other's behaviors. These activities may, in turn, generate trust toward each other and enhance cooperation and cohesion (Mathieu, Heffner, Goodwin, Salas, & Cannon-Bowers, 2000). In this way, team processes and the emergent states resulting from shared leadership may subsequently result in higher levels of team performance. Given the more direct or proximal relationship that is likely between shared leadership and team processes and emergent processes, we expect shared leadership to have a weaker relationship with more distal performance outcomes.

*Hypothesis 3:* The relationship between shared leadership and team processes and emergent states is stronger than the relationship between shared leadership and team performance.

**Work complexity as a contextual moderator.** The nature of the work that is pursued by a team may be relevant to the relationship between shared leadership and team effectiveness. Work complexity has been used as an important moderator in examining the relationship between team characteristics (e.g., goal setting) and group performance (Kleingeld, van Mierlo, & Arends, 2011). We specifically propose that more complex work will necessitate a higher degree of shared leadership to ensure team effectiveness. Work complexity in a team context can be defined in terms of the extent to which that work is highly knowledge-based and requires a large degree of information sharing and interdependent activities on the part of team members (Ensley et al., 2006; Hmieleski et al., 2012; Hoch & Dulebohn, 2013; Hoch & Kozlowski, in press).

Especially when a team faces much complexity, as well as associated uncertainty in achieving goals, there may be heightened demands for more than one individual to play the leader role (Day et al., 2004; Morgeson, DeRue, & Karam, 2010). In such a context, because of continuous novelty or disruption that is faced in the pursuit of goals, team members may need multiple members to be involved in information and perspective sharing and team learning (Hoch & Kozlowski, in press). Moreover, because of the potential frustrations of complex work, supportive leader behaviors on the

part of multiple team members may be necessary to help deal with frustrations or setbacks. In contrast, "tasks that are inflexible or uninteresting [i.e., less complex] will make shared leadership unnecessary or irrelevant" (Yammarino et al., 2012, p. 390). Arguments presented earlier pertaining to social identity and leader prototypicality may also be relevant here. Hogg et al. (2012) presented evidence suggesting that leader prototypicality is especially relevant to leader effectiveness under conditions of uncertainty or ambiguity. If shared leadership indeed broadens leader prototypicality, then we would expect shared leadership to be especially relevant to team effectiveness when work creates more uncertainty or ambiguity, that is, when work is more complex.

*Hypothesis 4:* The relationship between shared leadership and team effectiveness is stronger when work is more complex, compared with when it is less complex.

**Referent of measurement.** There are multiple ways to measure shared leadership, each of which has its pros and cons (Conger & Pearce, 2003). One common measure focuses on the influence of the team as a whole. This referent shift measurement procedure involves perceptions of the team as a whole (Chan, 1998), whereby all members evaluate the team as the entity of influence (Conger & Pearce, 2003). Consistent with this logic, researchers (e.g., Pearce & Sims, 2002) have used modified traditional leadership items (e.g., from the Multifactor Leadership Questionnaire) to assess the team as a whole, rather than individuals within the team. As another example, Pearce and Ensley (2004) developed an approach for assessing shared visionary leadership by asking individual team members the extent to which team members as a whole share in the development, creation, communication, and reinforcement of a common vision. For a respective team, responses are then aggregated across individual members to form a team-level, shared visionary leadership score. We should note that while assessing the shared influence in a team as a whole, this approach can smooth the differences in contributions of each individual member (Conger & Pearce, 2003).

A second method to assess shared leadership emphasizes the role of each individual team member. This method allows for an examination of the extent to which individuals are perceived to be involved in the sharing of leadership, as well as the distribution of leadership roles. For instance, the network density measures described above (Carson et al., 2007), which ask each member to rate each peer and then provides an average of members' scores to the team level, examine the overall or average amount of leadership across team members (Conger & Pearce, 2003). Third and related, shared leadership can also be self-assessed. For instance, Pearce and Ensley (2004) introduced a procedure to assess shared vision that involves each member rating him or herself, and these scores are then averaged to the team level.

As noted by multilevel researchers (e.g., Hofmann & Jones, 2005; Kozlowski & Klein, 2000), the main pitfall of focusing on the ratings of each individual team member is that we may fail to assess the potential impact targeted at the team as a whole. Compared with individual referent assessments, group referent approaches in general (i.e., for variables other than shared leadership) tend to exhibit stronger relationships with unit effectiveness, since both theory and measurement are aligned at the same level of analysis. For example, research indicates that the relationship of

justice climate and team effectiveness is stronger when the unit, rather than individual, referent is applied (Whitman et al., 2012). Similarly, it is possible that in terms of shared leadership, the group referent may capture and predict team effectiveness more accurately than individual-referent assessment. Moreover, the self-assessment of shared leadership may fall prey to common method variance. However, the evidence is not altogether clear. Accordingly, we explore whether the three alternative referents described above may serve as a moderator of the magnitude of the relationship between shared leadership and team effectiveness. We ask the following question:

*Research Question 1:* Do alternative referents for measuring shared leadership result in differential relationships with team effectiveness?

### Shared Versus Vertical Leadership

As an additional research issue pertaining to the content of leadership, we examine the relative roles of shared and vertical leadership. Because of the sheer volume of leadership that may be necessary in teams, and the limited ability of formal leaders to satisfy all leadership needs (e.g., because of competing time demands, physical distance from team members, and so forth), various authors have stressed the need for team members themselves to help assume leadership roles (e.g., Day et al., 2004; Pearce & Conger, 2003). However, this is not to necessarily downplay the relative role of hierarchically based, or vertical leadership. As noted by Hoch and Dulebohn (2013, p. 117), “shared leadership is not mutually exclusive to other leadership forms and behaviors, but can be engaged in simultaneously with other approaches such as vertical leadership.”

Thus, while we may witness both shared and vertical leadership in teams, it is not at all clear as to the relative importance of each. Indeed, it is possible that the truly important leadership activities in teams emanate from vertical leaders, since compared to team members, they may have more opportunities to exercise power and influence. On the other hand, vertical leadership may be insufficient, and shared leadership may provide a supplemental, unique explanation for variance in team effectiveness (Ensley et al., 2006; Pearce & Sims, 2002). Given the early state of our theoretical understanding of the relative importance of these two sources of leadership, we explore the following research question:

*Research Question 2:* Taking into account vertical leadership, does shared leadership account for unique variance in team effectiveness?

## Method

### Literature Search

In an attempt to identify all potential shared leadership studies, we searched the PsycINFO, ABI/Inform, Web of Science, and dissertation/thesis databases (as of July 2013) using the following keywords: shared leadership, team leadership, collective leadership, distributed leadership, overall leadership, integrated leadership, shared vision, and collective vision. We also searched the Society for Industrial and Organizational Psychology and the Academy of Management annual meeting programs from 2002 to

2013. We performed a Google Scholar search to look for unpublished working papers. Additionally, we solicited unpublished work from a number of leadership researchers, and we sent e-mails using multiple listservs of the Academy of Management divisions to seek working papers and unpublished data sets. Further, we manually checked the reference lists of all of the identified studies to ensure that we had captured all relevant articles. A total of 161 studies were found to be potentially relevant to the current topic.

### Criteria for Inclusion

There were three criteria guiding the selection process for inclusion of studies. First, research reports (e.g., published articles and working papers) should have offered enough information (e.g., zero-order correlations between shared leadership and team effectiveness) to allow effect size computation. Second, research reports needed to provide team-level shared leadership scores. We excluded studies that only examined the effects of shared leadership at the individual level of analysis (e.g., Avolio et al., 1996; Davis, 1995). We also excluded studies that captured members' teamwork behaviors, despite their usage of “shared leadership” to label such variables (e.g., Muethel, Gehrlein, & Hoegl, 2012). Third, we only included studies with adult samples (employees or adult students) and excluded experimental studies in which shared leadership was manipulated by the researchers (e.g., Belau, Bell, & Briggs, 2008). It is notable that we initially identified several studies from the education literature, in which the total leadership exhibited across several hierarchical levels is examined (e.g., Leithwood & Jantzi, 2000; Leithwood & Mascal, 2008). We ultimately excluded those studies, since their measurement was confounded by vertical leadership influence (e.g., on the part of school principals and administrators), rather than having an exclusive focus on shared leadership among peers.

To ensure coding accuracy, two authors were trained in terms of shared leadership and team effectiveness coding rules. After the training, the authors initially coded five articles independently and then discussed them to ensure that they interpreted the variables the same way. Subsequently, each author independently coded the remaining articles. The initial interrater agreement rates were high, ranging from 98% to 100% for the coding of shared leadership content, team effectiveness, measurement referents, effect sizes, sample size and other characteristics, as well as coefficient alphas and agreement indices (i.e.,  $r_{wg(j)}$ , intraclass correlations [ICC<sub>1</sub>, ICC<sub>2</sub>] of the primary studies). The initial interrater agreement was somewhat lower for coding work complexity (89% agreement, Cohen's kappa = .82,  $p < .001$ ) due to its heightened level of subjective judgment based on task descriptions of the primary studies. All of the remaining discrepancies were subsequently resolved through double-checking the primary studies and discussions between the two coders. This process yielded 40 studies in total (with 42 independent samples; see Appendix A and Appendix B for the meta-analyzed and the examined but excluded studies, respectively), covering 3,439 teams and 32,616 individuals, with an average size of 9.48 members per team.

### Variables Coded

**Content of shared leadership.** There are various types of shared leadership represented in these studies in terms of their content. In Table 1, we list how each measure of shared leadership



Table 1  
*Categorization of Shared Leadership Content*

| Measure of shared leadership          | Indicators                                                                                                                                                                                                                                            |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Shared traditional leadership         | Transactional leadership, initiating structure, consideration behaviors, management by exception, directive, aversive (reverse), planning and organizing, problem solving, support and consideration, participative leadership, supportive leadership |
| Shared new-genre leadership           | Visionary leadership, transformational leadership, charismatic leadership, empowering leadership, authentic leadership                                                                                                                                |
| Cumulative, overall shared leadership | Emergent leadership aggregated to the team level                                                                                                                                                                                                      |

from the primary studies fits into our categorization. For shared traditional leadership, we include the Ohio State leadership measures (i.e., initiating structure and consideration), transactional leadership, participative leadership, and so forth. As an example, for shared initiating structure and consideration, sample items are “to what extent do crew members emphasize the meeting of deadlines?” and “to what extent do crew members do little things to make it pleasant to be an employee in the restaurant?” (Ziegert, 2005). For shared transactional leadership, sample items include “members of my team allow performance to fall below minimum standards before trying to make improvements” (Avolio, Sivasubramaniam, Murry, Jung, & Garger, 2003).

Shared new-genre leadership includes shared visionary, transformational, authentic, and so forth, leadership. A sample item for shared transformational leadership is “members of my team instill pride in being associated with each other” (Avolio et al., 2003). As another example, based on the work of Pearce and Ensley (2004), a sample item of shared visionary leadership is “because of my team members, I have a clear vision of our team’s purpose” (Pearce & Ensley, 2004). For cumulative, overall shared leadership, one example is from the work of Carson et al. (2007), who used a social network method of measuring shared leadership. They asked respondents “to what degree does your team rely on this individual for leadership?” and then used social network density scores to assess the distribution of emergent, shared leadership with a team.

**Team effectiveness.** As discussed above, we excluded the withdrawal category from Roberson and Colquitt’s (2005) framework of team criteria due to the lack of studies examining withdrawal. We distinguished between subjective and objective performance. As such, we have four categories of team effectiveness: (a) attitudinal outcomes, (b) behavioral processes and emergent states, (c) subjective performance, and (d) objective performance. Attitudinal outcomes are largely affective in nature, including team satisfaction, team commitment, team identification, and emotional conflicts (reversed). Behavioral processes and emergent states include team cohesion and coordination, helping, and so forth. Subjective performance involves performance ratings provided by team leaders, members themselves, or external, higher level managers. Objective measures of performance include indicators such as actual sales and productivity.

**Work complexity.** Based on the primary studies’ descriptions of team tasks, we coded the complexity of work in a given sample using a 5-point Likert-type scale. The least complex work (= 1) involved a low level of knowledge use and requires little creative thinking and low information sharing among members (i.e., members can finish the work independently). The most complex work

(= 5) involved a high level of knowledge use and high creative thinking and extensive information sharing among members (i.e., members are highly interdependent in the pursuit of a collective goal; Crawford & LePine, 2013; Gully, Incalcaterra, Joshi, & Beaubien, 2002; Langfred, 2007).

**Referent of assessment.** Shared leadership is a group-level construct. However, as noted earlier, researchers have measured it using different referents. Accordingly, we evaluated three potential referents: (a) team as the referent, (b) each peer as the referent (i.e., network density approach), and (c) each member rating him or herself.

**Vertical leadership.** We also coded the correlations between shared leadership and vertical leadership that is exhibited by the formal, designated leaders in the primary studies. We separated the content of vertical leadership into traditional versus new-genre vertical leadership in a similar manner as we did for shared leadership. This separation helped us to examine the incremental influence of shared traditional and shared new-genre leadership over and above the corresponding forms of vertical leadership. Among the 42 independent samples that we identified, no study had measured vertical leadership in terms of how we have defined the cumulative, overall form, so we excluded this type for vertical leadership coding.

**Publication status and team type.** We coded a primary study as published if it is a journal article or a book chapter. The study is considered as unpublished if it is an unpublished dissertation, a conference paper, a working paper, or if the data are not published in any form of formal publications. Team type was coded such that intact work teams and masters of business administration (MBA) student teams engaging in real-life consulting projects are considered work teams. Undergraduate student teams working on course projects or other assigned tasks were coded as student teams.

## Meta-Analytical Procedures

We followed the random effects, meta-analytic procedures described by Hunter and Schmidt (2004). We report both the sample-size weighted, mean observed correlations and the correlations that have been corrected for measurement errors (but not for range restriction). Specifically, attitudinal outcomes and behavioral/emergent states were corrected for unreliability based on reported ICC<sub>2</sub>. ICC<sub>2</sub> indicates the reliability of team-level, mean scores (Bliese, 2000), and it has been used consistently in prior meta-analyses as the basis for correcting unreliability of team-level constructs (e.g., Seibert, Wang, & Courtright, 2011; Whitman et al., 2012). When subjective performance was rated by a single respondent (i.e., an external manager), we used the alpha coefficient

cient to correct for unreliability in this outcome variable. When team members provided ratings on team subjective performance, we used  $ICC_2$  instead. For the small proportion of studies that did not report a reliability estimate for the outcome variables, we used the average of available reliabilities (average reliabilities are attitudinal outcomes = .81, behavioral processes and emergent states = .83, and subjective performance = .89). Objective performance variables were not corrected for unreliability due to the lack of information provided in primary studies. For shared leadership measures, the majority of primary studies did not report  $ICC_2$  values (only 14 out of the 42 samples reported  $ICC_2$ ). Therefore, for these studies without  $ICC_2$  information, we chose a value of .80 for correcting unreliability of shared leadership. This value is comparable to the  $ICC_2$  values used in prior meta-analyses on team-level constructs (e.g., the mean  $ICC_2$  values was .75 in the justice climate meta-analysis conducted by Whitman et al., 2012).

When multiple correlations of a relationship were available from a single sample, we combined them into one correlation using the formula for composites (Hunter & Schmidt, 2004). As such, the effect sizes in our meta-analysis were independent. Consistent with Hunter and Schmidt (2004), we created subsets of correlations to test categorical moderators. Moderation by a categorical moderator is inferred based on between-subset differences in the mean estimated effect sizes and their 95% confidence intervals. To examine the continuous moderator (work complexity), we followed the recommendations by Steel and Kammeyer-Mueller (2002) and used weighted least square (WLS) regression. In the meta-analytic, WLS regression analyses, effect sizes were given inverse variance weights based on their sample size (Hedges & Olkin, 1985). We used the mixed-effect models in the metafor package of the R program (Viechtbauer, 2010). Although our sample size ( $k = 42$ ) does not afford us to control for other contextual variables, we performed the WLS regression with and without controlling for team type (work team = 1 and student team = 0). The results are nearly identical, and we only report the results without the control variable. To examine Research Question 2, we used path analysis based on meta-analytically derived correlation matrices (Viswesvaran & Ones, 1995). We supplemented our newly calculated true-score correlations with effect sizes from prior meta-analyses (Judge et al., 2004; Judge & Piccolo, 2004). More details are provided in the Results section.

**Outlier detection and publication bias.** As a supplemental analysis, we used the sample-adjusted meta-analytic deviancy

(SAMD; Hufcutt & Arthur, 1995) statistic but failed to identify any potential outliers on the basis of sample size and magnitude of coefficients. Furthermore, to test for potential publication biases, we first conducted subgroup analyses and examined the estimated effect sizes for published versus unpublished data. There are 19 unpublished samples out of the total of 42 unique samples that we identified. Second, we created contour-enhanced funnel plots using the “confunnel” command in Stata (Kepes, Banks, McDaniel, & Whetzel, 2012; Palmer, Peters, Sutton, & Moreno, 2008). The distribution of samples was symmetric in the plot, therefore indicating that publication bias is likely to be absent (Sterne & Harbord, 2004).

## Results

Table 2 shows the overall relationship between shared leadership and team effectiveness. The uncorrected sample-size weighted mean correlation between shared leadership and team effectiveness was .29. After we corrected for measurement errors in both shared leadership and team effectiveness, the overall mean correlation became  $\rho = .34$ . Both correlations suggest a moderately strong association between shared leadership and team effectiveness. Additionally, neither the 80% credibility interval nor the 95% confidence interval includes zero, indicating the generalizability of the positive relationship across studies. Taken together, our results supported Hypothesis 1.

We also report the results of supplemental analyses in Table 2. Regarding publication status, the corrected correlations for published and unpublished studies are similar ( $\rho = .35$  and .32, respectively), and their 95% confidence intervals overlap with each other. This indicates that publication bias in our analyzed studies is at a minimum. Similarly, team type does not influence the overall relationship between shared leadership and team effectiveness. Although work teams ( $\rho = .35$ ) and student teams ( $\rho = .28$ ) show somewhat different correlations, their 95% confidence intervals overlap with each other.

Table 3 presents the results for testing Hypotheses 2a, 2b, and 3 and Research Question 1. Regarding Hypothesis 2a, the corrected correlations suggest that the relationship between shared traditional leadership and team effectiveness ( $\rho = .18$ ) is weaker than that between shared new-genre leadership and team effectiveness ( $\rho = .34$ ). The 95% confidence intervals of these two correlations do not overlap with each other, suggesting that these two relation-

Table 2  
*Meta-Analytic Relationships Between Shared Leadership and Team Effectiveness*

| Variable             | <i>k</i> | <i>N</i> | $\bar{r}$ | $\rho$ | $SD_{\rho}$ | %Var  | Lower CV | Upper CV | Lower CI | Upper CI |
|----------------------|----------|----------|-----------|--------|-------------|-------|----------|----------|----------|----------|
| Overall relationship | 42       | 3,439    | .29       | .34    | .10         | 60.0  | .21      | .46      | .29      | .38      |
| Publication status   |          |          |           |        |             |       |          |          |          |          |
| Published studies    | 23       | 1,537    | .30       | .35    | .00         | 100.0 | .35      | .35      | .31      | .40      |
| Unpublished studies  | 19       | 1,902    | .28       | .32    | .13         | 40.2  | .16      | .49      | .25      | .39      |
| Team type            |          |          |           |        |             |       |          |          |          |          |
| Work teams           | 31       | 2,913    | .30       | .35    | .11         | 49.5  | .20      | .49      | .30      | .40      |
| Student teams        | 11       | 526      | .24       | .28    | .00         | 100.0 | .28      | .28      | .20      | .36      |

*Note.* *k* = number of studies cumulated; *N* = cumulative sample size (number of teams);  $\bar{r}$  = sample-size weighted correlation, not corrected for measurement errors;  $\rho$  = sample-size weighted correlation corrected for measurement errors;  $SD_{\rho}$  = standard deviation of  $\rho$ ; %Var = percentage of variance attributable to statistical artifacts; CV = 80% credibility interval of  $\rho$ ; CI = 95% confidence interval.



Table 3  
Categorical Moderator Analyses for Shared Leadership and Team Effectiveness

| Variable                                 | <i>k</i> | <i>N</i> | $\bar{r}$ | $\rho$ | $SD_{\rho}$ | %Var  | Lower CV | Upper CV | Lower CI | Upper CI |
|------------------------------------------|----------|----------|-----------|--------|-------------|-------|----------|----------|----------|----------|
| Content of shared leadership             |          |          |           |        |             |       |          |          |          |          |
| Shared traditional leadership            | 11       | 684      | .16       | .18    | .00         | 100.0 | .18      | .18      | .11      | .25      |
| Shared new-genre leadership              | 25       | 2,162    | .30       | .34    | .11         | 52.6  | .20      | .48      | .28      | .40      |
| Cumulative, overall shared leadership    | 14       | 1,170    | .30       | .35    | .12         | 49.3  | .20      | .50      | .27      | .43      |
| Criteria types                           |          |          |           |        |             |       |          |          |          |          |
| Attitudinal outcomes                     | 12       | 944      | .39       | .45    | .12         | 47.5  | .30      | .61      | .37      | .54      |
| Behavioral processes and emergent states | 27       | 2,359    | .37       | .44    | .14         | 39.6  | .26      | .61      | .37      | .50      |
| Different source from shared leadership  | 6        | 532      | .27       | .31    | .00         | 100.0 | .31      | .31      | .23      | .38      |
| Same source as shared leadership         | 21       | 1,827    | .40       | .46    | .13         | 39.7  | .29      | .62      | .39      | .53      |
| Subjective performance                   | 19       | 1,866    | .22       | .25    | .11         | 50.9  | .11      | .39      | .18      | .32      |
| Different source from shared leadership  | 15       | 1,583    | .20       | .24    | .09         | 61.0  | .13      | .35      | .17      | .30      |
| Same source as shared leadership         | 6        | 421      | .28       | .31    | .14         | 42.4  | .12      | .49      | .16      | .45      |
| Objective performance                    | 23       | 1,630    | .16       | .18    | .14         | 48.4  | .01      | .35      | .11      | .25      |
| Referent of measurement                  |          |          |           |        |             |       |          |          |          |          |
| Team as the referent                     | 24       | 1,875    | .27       | .31    | .08         | 71.3  | .21      | .41      | .26      | .37      |
| Each peer as the referent                | 14       | 1,171    | .31       | .37    | .11         | 54.8  | .23      | .50      | .29      | .44      |
| Each member rating him/herself           | 6        | 681      | .31       | .37    | .09         | 54.9  | .25      | .49      | .27      | .47      |

Note. *k* = number of studies cumulated; *N* = cumulative sample size (number of teams);  $\bar{r}$  = sample-size weighted correlation, not corrected for measurement errors;  $\rho$  = sample-size weighted correlation corrected for measurement errors;  $SD_{\rho}$  = standard deviation of  $\rho$ ; %Var = percentage of variance attributable to statistical artifacts; CV = 80% credibility interval of  $\rho$ ; CI = 95% confidence interval. For criteria types, attitudinal outcomes are all from the same source as shared leadership, and objective performance measures are all from a different source from that of shared leadership.

ships are significantly different. Thus, Hypothesis 2a was supported. In addition, cumulative, overall shared leadership ( $\rho = .35$ ) is more predictive of team effectiveness compared to shared traditional leadership ( $\rho = .18$ ). The 95% confidence intervals (CIs) of cumulative, overall shared leadership (CI = .27, .43) and shared traditional leadership (CI = .11, .25) do not overlap, suggesting that the magnitude of the associations differed significantly. Hypothesis 2b was supported.

For Hypothesis 3, we predicted that shared leadership was differentially related to various team effectiveness criteria. Table 3 shows the relationships between shared leadership and the four types of criteria (i.e., attitudinal outcomes, behavioral processes and emergent states, subjective performance, and objective performance). Shared leadership is more strongly related to attitudinal outcomes and behavioral processes and emergent states ( $\rho = .45$  and .44, respectively), compared with subjective and objective outcomes ( $\rho = .25$  and .18, respectively). As shown in Table 3, the confidence intervals for attitudinal and behavioral processes and emergent states are similar to each other, and neither of them overlaps with the confidence intervals for subjective and objective performance.

Compared with objective performance indices, subjective measures of team performance, as well as team member-rated behavioral processes and emergent states, can be more easily impacted by a number of factors such as common methods variance, social desirability, and self-deceptions (Buckley, Cote, & Comstock, 1990; Campbell & Fiske, 1959; Donaldson & Grant-Vallone, 2002). For instance, members' ratings of team performance would be more closely related to shared leadership if those ratings are completed by the same source (Lowe, Kroeck, & Sivasubramaniam, 1996). As such, it is possible that the association between shared leadership and effectiveness criteria is reduced when researchers use more objective outcome measures (e.g., productivity and actual sales).

Therefore, to further investigate the role played by common method variance, we split the studies into two groups (i.e., same versus different sources for the shared leadership and team effectiveness measures). After the split, the effect size for behavioral processes and emergent states measured from a different source ( $\rho = .31$ ,  $k = 6$ ) is still greater than that for subjective performance measured from a different source ( $\rho = .24$ ,  $k = 15$ ). However, with the reduced *k*, their 95% confidence intervals now overlap with each other (CI = .23 to .38 and .17 to .30, respectively). Similarly, the effect size for behavioral processes and emergent states measured from the same source ( $\rho = .46$ ,  $k = 21$ ) is greater than that for subjective performance measured from the same source ( $\rho = .31$ ,  $k = 6$ ). With the reduced *k*, their 95% confidence intervals also overlap with each other (CI = .39 to .53 and .16 to .45, respectively). Taken together, these results provide some cautionary support for Hypothesis 3. That is, they suggest that the magnitudes of the associations between shared leadership and various team effectiveness criteria differ, although to some extent this finding needs to be qualified by the potential influence of common method variance between shared leadership and the effectiveness measures.

Hypothesis 4 suggested a positive relationship between work complexity and the effect size of shared leadership with team effectiveness. Table 4 shows the WLS regression results. With 42 independent samples, work complexity is positively related to the effect size ( $b = .04$ ,  $SE = .02$ , one-tailed  $p < .05$ , 90% CI = .005, .07). Although not reported in this table, the results remain the same after we controlled for team type (work team = 1, student team = 0). Thus, Hypothesis 4 was supported.

Research Question 1 pertains to the referent of measurement. As shown in Table 3, the shared leadership—team effectiveness relationships are similar among the various types of measurement referents. The 95% confidence intervals of the three types of measurement referents overlap with each other. These findings

Table 4

*Weighted Least Square Regression Analysis on Work Complexity as a Continuous Moderator*

| Variable        | Shared leadership–team effectiveness |           |             |
|-----------------|--------------------------------------|-----------|-------------|
|                 | <i>B</i>                             | <i>SE</i> | 90% CI      |
| Constant        | .25***                               | .07       | (.13, .37)  |
| Work complexity | .04*                                 | .02       | (.005, .07) |

*Note.* Unstandardized weighted least square regression coefficients are presented.  $R^2 = .12$ ;  $F(1, 40) = 5.65$ ,  $p = .02$ ;  $k = 42$ . CI = confidence interval;  $k$  = the total number of effect sizes.

\*  $p < .05$  (one-tailed test). \*\*\*  $p < .001$ .

indicate that the relationship between shared leadership and team effectiveness did not differ across the various referents. In other words, alternative referents appear to not result in significantly differential relationships between shared leadership and team effectiveness.

Research Question 2 pertains to the unique effect of shared leadership on team effectiveness after taking into account vertical leadership. We followed the procedures suggested by Viswesvaran and Ones (1995). Table 5 provides the correlation matrices used in our analysis, and we examined traditional and new-genre leadership separately. For each of these two types, the true score correlations between shared and vertical leadership, and between shared leadership and team performance, come from our current meta-analyses. We obtained the true-score correlations between vertical leadership and team performance (cells indicated by <sup>a</sup> or <sup>ab</sup>) from Judge et al. (2004) and Judge and Piccolo (2004). The former study provided meta-analytic results for traditional forms of leadership, while the latter study provided meta-analytic findings for transformational leadership (which is highly representative of new-genre leadership), as well for transactional leadership (another form of traditional vertical leadership). Because these prior meta-analyses combined subjective and objective team performance in their results, we similarly combined the two types of team performance in our calculation of other cells in the correlation matrix.

Table 6 provides the results of path modeling analysis. The sample size for each model was the harmonic mean of the sample sizes in all of the cells of the meta-analyzed correlation matrix.

Table 5

*Meta-Analytic Correlation Matrix Among Shared Leadership, Vertical Leadership, and Team Performance*

| Variable                           | 1                                                         | 2                                                             | 3 |
|------------------------------------|-----------------------------------------------------------|---------------------------------------------------------------|---|
| Traditional leadership             |                                                           |                                                               |   |
| 1. Shared traditional leadership   | —                                                         |                                                               |   |
| 2. Vertical traditional leadership | $\rho = .38$ ( $k = 5$ , $N = 358$ , $CI = .27, .50$ )    | —                                                             |   |
| 3. Team performance                | $\rho = .16$ ( $k = 10$ , $N = 639$ , $CI = .07, .23$ )   | $\rho = .23^{ab}$ ( $k = 70$ , $N = 7,314$ , $CI$ excludes 0) | — |
| New-genre leadership               |                                                           |                                                               |   |
| 1. Shared new-genre leadership     | —                                                         |                                                               |   |
| 2. Vertical new-genre leadership   | $\rho = .64$ ( $k = 11$ , $N = 1,124$ , $CI = .57, .72$ ) | —                                                             |   |
| 3. Team performance                | $\rho = .22$ ( $k = 22$ , $N = 2,060$ , $CI = .16, .29$ ) | $\rho = .26^a$ ( $k = 41$ , $N = 6,197$ , $CI$ excludes 0)    | — |

*Note.*  $\rho$  = true correlations;  $k$  = number of independent samples;  $N$  = accumulated total number of teams from primary studies;  $CI$  = 95% confidence interval. Each cell in the matrix contains a meta-analyzed correlation either from the current meta-analysis or from prior meta-analyses (denoted by <sup>a</sup> or <sup>ab</sup>).

<sup>a</sup> Indicates the  $\rho$  is from Judge and Piccolo's (2004, p. 760) Table 2, that is, relationship between transformational leadership and team performance. <sup>ab</sup> Indicates the  $\rho$  is from a combination of results from Judge and Piccolo (2004, p. 760) and Judge, Piccolo, and Ilies (2004, p. 40), that is, the effect sizes for contingent reward, initiating structure, and consideration were combined (weighted sum by respective sample size).

Following prior research (e.g., Chiaburu, Oh, Berry, Li, & Gardner, 2011), we report multiple  $R$ ,  $R^2$ , and the change in multiple  $R$  associated with adding a predictor to the model. The changes in multiple  $R$  show the incremental predictive power of the variable above and beyond other predictors in the model. As Table 6 reveals, shared traditional leadership significantly predicts team performance ( $\beta = .09$ ,  $p < .05$ ) after controlling for vertical traditional leadership, and its  $\Delta R$  over vertical leadership is .01 ( $p < .05$ ). Similar results were obtained for shared new-genre leadership, with standardized coefficient  $\beta = .09$  ( $p < .01$ ) and  $\Delta R$  of .01 ( $p < .01$ ). These findings indicate that shared leadership accounts for unique variance in team effectiveness after taking into account vertical leadership.

## Discussion

We conducted the current meta-analytic research to systematically examine the association between shared leadership and team effectiveness. The results confirmed our prediction that there is a moderately strong, positive relationship between shared leadership and team effectiveness (overall  $\rho = .34$ ,  $k = 42$ ). This finding confirms the importance of examining shared leadership in teams (Denis et al., 2012; Yammarino et al., 2012). That is, it is consistent with Pearce and Conger's (2003) perspective that shared leadership is essential to team goal achievement.

## Theoretical Implications

To provide a more nuanced understanding, we investigated how the relationship differs with regard to several potential moderators. First, shared new-genre leadership exhibits a stronger relationship with team effectiveness compared with that of shared traditional leadership. This finding suggests that new-genre leadership, which emphasizes change and development (Contractor et al., 2012), is more beneficial as a shared leadership function, compared with the more traditional forms of leadership that may rely on exchanges or the maintenance of the status quo. It could be that the sharing of leadership functions that are oriented toward change (e.g., shared visionary leadership) is especially relevant to the achievement of team outcomes. In contrast, it is not so essential for multiple team members to be involved in more traditional leadership functions.

Table 6  
*Shared Leadership and Vertical Leadership in Predicting Team Performance*

| Variable                                 | Team performance |
|------------------------------------------|------------------|
| Traditional leadership                   |                  |
| Vertical traditional leadership          | .20***           |
| Shared traditional leadership            | .09*             |
| Harmonic sample size                     | <i>N</i> = 667   |
| Total <i>R</i> ( <i>R</i> <sup>2</sup> ) | .24 (.06)        |
| Δ <i>R</i> over shared leadership        | .08***           |
| Δ <i>R</i> over vertical leadership      | .01*             |
| New-genre leadership                     |                  |
| Vertical new-genre leadership            | .20***           |
| Shared new-genre leadership              | .09**            |
| Harmonic sample size                     | <i>N</i> = 1,953 |
| Total <i>R</i> ( <i>R</i> <sup>2</sup> ) | .27 (.07)        |
| Δ <i>R</i> over shared leadership        | .05***           |
| Δ <i>R</i> over vertical leadership      | .01**            |

Note. Standardized coefficients are reported.

\* *p* < .05. \*\* *p* < .01. \*\*\* *p* < .001.

Given the increasing importance of ongoing change in teams and organizations (Beer & Nohria, 2000), our findings serve to highlight the possibility that shared leadership may need to accompany change and development efforts.

Moreover, shared new-genre leadership and cumulative, overall shared leadership exhibit similar validity in terms of team effectiveness. This finding further supports the leadership prototypicality theory such that people have their ideal leadership characteristics in mind (Lord et al., 1984) when evaluating overall leadership. According to this theory, leader prototypes appear to be consistent with the new-genre leadership as characterized here by, for example, visionary and authentic behaviors (Avolio, 2007). Therefore, shared cumulative, overall leadership demonstrates a similarly strong relationship with team effectiveness as does shared new-genre leadership. It might be informative for future research to compare the effects of shared leadership in terms of leader prototypes based on implicit leadership theory, versus social identity. Hogg et al. (2012) suggested that as team membership becomes more salient, implicit leadership theories will become less relevant to leader evaluations and effectiveness.

Second, although shared leadership significantly relates to team effectiveness, in terms of different types of effectiveness criteria, the relationship appears to vary. Specifically, shared leadership is more related to attitudinal and behavioral outcomes, compared with performance measures (both subjective and objective ones). This is consistent with Yammarino et al.'s (2012) view that shared leadership, as an aspect of team process, can also be an input to other team processes or emergent properties of the team. Thus, by sharing leadership roles in a team, members work toward a common goal and lead each other's behaviors. These activities may, in turn, generate trust toward each other and enhance cooperation and cohesion (Mathieu et al., 2000). In this way, team processes and the emergent states resulting from shared leadership may subsequently result in higher levels of team performance. It should be noted that the overall pattern of our findings were largely replicated in a more nuanced analysis that attempted to delineate same-source from different-source findings. However, because of reduced numbers of

studies involved in these analyses, overlapping confidence intervals precluded definitive conclusions when comparison studies were based on either different-source or same-source data.

Third, we found that the complexity of the work performed by teams in the primary studies that were meta-analyzed served to moderate the relationship between shared leadership and outcomes. When work is more knowledge-based and interdependent, there is a stronger relationship between shared leadership and outcomes. This finding provides further insight into the circumstances under which shared leadership might provide the most utility. Organizations are increasingly relying upon knowledge-based teams that engage in complex work. Perhaps the heightened degree of leader prototypicality that may accompany shared leadership is especially relevant to the uncertainty or ambiguity associated with complex work (Hogg et al., 2012). Future research on shared leadership might focus on teams involved in complex work, since they operate in a context that is both theoretically and practically relevant.

Our results also suggest that the referent of measurement has no effect in terms of the prediction of outcomes. The confidence intervals of the three types of referents overlap with each other. These findings indicate that regardless of which measurement approach researchers adopt, the overall positive relationship between shared leadership and team effectiveness is likely to stay essentially the same. Previous research has discussed the application of different methods for measuring a unit-level construct to capture the multiple facets of the construct (Chan, 1998). Since shared leadership has not been shown as a multifaceted construct (Ziegert, 2005), our meta-analysis thus suggests that the referents used do not alter the relationship between shared leadership and team effectiveness as long as the measurement procedure can capture the shared characteristics of leadership in a team.

We further demonstrated that shared leadership has a unique effect when vertical leadership is taken into account. This finding suggests the value-added potential of shared leadership, although it may actually represent an underestimate of relative effects. Specifically, although our results appear to be weighted more in the direction of the relevance of vertical leadership, there may be more of a tendency to make attributions toward a singular, vertical leader based on observed performance. In other words, the relatively stronger effects that we reported for vertical leadership may be at least somewhat due to leadership attributions that are made based on performance that people observe regarding vertical leaders (Agle, Nagarajan, Sonnenfeld, & Srinivasan, 2006). Such attributions may not be as prominent when leadership is dispersed and shared, and thus, performance effects for shared leadership are more likely to be actual, rather than attributional in nature. We should also note that our analyses were based on estimates from prior meta-analyses involving vertical leadership. Some existing research has directly examined the relative importance of vertical and shared leadership in terms of team effectiveness (e.g., Ensley et al., 2006; Pearce & Sims, 2002). Their results support the notion that although both types of leadership can enhance team performance, shared leadership may actually explain more variance in the prediction of team effectiveness when both sources of leadership are assessed in the same study.



## Limitations and Future Research

Although we believe that our study helps advance the shared leadership literature in various ways, it also has limitations. The most important limitation is the relatively moderate number of studies available for meta-analytic purposes. However, with the data that are currently available, we were able to provide insights regarding the effects of alternative content, types of outcomes, contextual conditions, and referents of shared leadership. An additional limitation is our inability to offer strong causal inferences. The primary studies that we meta-analyzed were not experimental studies; therefore, we need to use caution when considering the causal directions between shared leadership and team effectiveness. Furthermore, only a few studies that were included in our meta-analysis had a time lag between the measurement of shared leadership and team effectiveness.

We suggest several potential avenues for future research. First, although we examined alternative referents for shared leadership, in each instance, the result is scoring that reflects the average or sheer amount in a respective team. However, the more qualitatively based distribution of leader roles is not examined in prior research. Thus, one potential research question is whether there is an optimal mix of leader roles within a team (Yukl, 1999). For example, should one team member emphasize visionary leadership, while another concentrates on task direction, while yet another takes primary responsibility for taking an interpersonally supportive role? Stated another way, there may be ideal configurations of shared leadership that prior conceptualizations and operationalizations have not taken into account.

As a related issue, the notion of “shared” leadership can potentially be confusing, especially with regard to leadership vision. Pearce and Ensley’s (2004) conceptualization and measurement of shared leadership actually appears to get at some combination of the amount of visionary leadership behavior that is demonstrated among team members, as well as actual agreement as to the nature of a vision. However, their approach could cause confusion in our understanding of what shared visionary leadership is all about. As a way to help address whether the Pearce and Ensley (2004) approach might possibly have affected our findings, we conducted supplemental analyses comparing results of studies that have used Pearce and Ensley’s (2004) measure, versus those that use other measures of shared leadership. Our findings suggested no discernible differences. Nevertheless, to avoid confusion, future researchers may want to clearly separate shared visionary behavior from shared vision.

Second, like other researchers (e.g., Morgeson et al., 2010; Yammarino et al., 2012), we suggest additional research to better understand the coexistence of these two sources of leadership within a team. In line with the results of Pearce and Sims (2002), we found significant, unique effects for both shared and vertical leadership. We propose that further research be conducted to ascertain the conditions under which shared leadership may be especially important. For example, our findings would suggest that shared leadership may be especially important in terms of showing unique effects in complex work environments.

Future research might also address the interplay of vertical and shared leadership. For example, shared leadership may partially mediate the relationship between vertical leadership and outcomes. Through the transcendence of self-interests and articulation of compelling vision, transformational leaders may foster a collective sense of identity in teams, as well as the motivation among team

members to exercise collective influence (Bass & Bass, 2008; Brown & Treviño, 2006). In short, “transformational leadership involves the process whereby leaders develop followers into leaders” (Avolio, 2011, p. 51). Accordingly, we might expect vertical, transformational leadership to be associated with a higher degree of shared leadership, which, in turn, is predictive of team outcomes. At the same time, we recognize that our findings also suggest unique and direct effects on the part of vertical leadership.

Moreover, in addition to shared leadership potentially mediating the relationship between vertical leadership and team outcomes, interaction effects are also possible. Using leader vision as an example, vertical leadership may not only affect the extent to which team members share in vision formation and acceptance. Shared visionary leadership may also work together, or interact, with vertical leader vision to help achieve outcomes. Such thinking follows from the work of Kohles, Bligh, and Carsten (2012), who argued that the role of followers in the formation and implementation of vision is equal to, if not greater, than that of vertical leaders. Indeed, their work would suggest that the vision of vertical leaders and team members might work best in tandem or partnership.

Third, in the current study, we focused on theoretically derived moderators of shared leadership–team effectiveness relationships, but our set of moderators is not exhaustive. For example, respective team members may need to have requisite expertise, information, or individual performance potential in order to effectively engage in shared leadership (Mumford, Friedrich, Vessey, & Ruark, 2012; Shin & Zhou, 2007). Carson et al. (2007, p. 1230) discussed “the distribution of task competence” on the part of individual team members as a potential moderator in the relationship between shared leadership and team performance.

Fourth, time may be an important factor in the development of shared leadership (Shamir, 2011). Aside from the causal inferences that can be gleaned from longitudinal data, time also plays an important role in team development. As time goes forward, team members may be more familiar with their tasks and know each other’s expertise (Mathieu et al., 2000), which can enhance the sharedness of leadership in a team. Increasing sharedness can further promote team processes and performance. For example, the common vision becomes clearer. Further, team members get a better understanding of how to collaborate with each other in an effective manner. Thus, shared leadership might be more beneficial to team effectiveness if members stay in their respective team over a relatively longer period of time (Mumford et al., 2012).

Fifth, we should note that for all studies that we identified in the literature assessing cumulative, overall shared leadership, each peer served as the referent. In other words, these studies uniformly used the social network density approach that we described earlier. However, it is also possible for future researchers to use the team as a whole as the referent when assessing shared leadership in such a generic manner. For example, team members could be asked the extent to which, in general, other team members share in the leadership role.

## Conclusion

Recent research has started to emphasize the importance of “we” in leadership processes (Yammarino et al., 2012). The current study lends substantial empirical support to a positive relationship between shared leadership and team effectiveness. Our results indicate that the link between shared leadership and team effectiveness is stronger

when members share new-genre leadership and cumulative, overall leadership in a team, compared with sharing more traditional leadership behaviors. In addition, we can tentatively conclude that shared leadership connects more strongly with attitudinal outcomes and behavioral processes and team emergent states than with actual team performance. Further, the effects of shared leadership are more strongly observed when the work of team members is more complex. We also show that shared leadership has unique effects on team outcomes above and beyond vertical leadership. Future research should build on our findings to move the study of shared, and more broadly plural, forms of leadership forward.

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# Appendix A

## Coding Information for Samples Included in the Meta-Analyses

| Study                          | Sample size<br>(no. of teams) | Content<br>of shared<br>leadership | Overall<br>effect<br>size | Effectiveness<br>measure | Measurement | Publication<br>status | Team<br>type | Include<br>vertical<br>leadership |
|--------------------------------|-------------------------------|------------------------------------|---------------------------|--------------------------|-------------|-----------------------|--------------|-----------------------------------|
| Balthazard et al. (2008)       | 88                            | N                                  | .36                       | B, OP                    | TE          | P                     | S            |                                   |
| Balthazard et al. (2010)       | 118                           | N                                  | .39                       | B, OP                    | TE          | U                     | S            |                                   |
| Bergman et al. (2012)          | 45                            | T, N, C                            | .14                       | A, B                     | TE          | P                     | S            |                                   |
| Berson et al. (2013)           | 103                           | N                                  | .30                       | B, OP                    | IN          | U                     | W            | V                                 |
| Boies et al. (2010)            | 49                            | N                                  | .32                       | A, B, OP                 | TE          | P                     | S            |                                   |
| Carson et al. (2007)           | 56                            | C                                  | .46                       | OP                       | PE          | P                     | W            | V                                 |
| Cashman (2008)                 | 31                            | N                                  | .76                       | B                        | TE          | U                     | W            | V                                 |
| DeRue et al. (2012)            | 171                           | C                                  | .33                       | B, SP                    | PE          | U                     | W            |                                   |
| Drescher et al. (2012)         | 174                           | A                                  | .23                       | A, B, OP                 | IN          | U                     | W            |                                   |
| Ensley et al. (2006; Sample 1) | 66                            | T, N                               | .22                       | OP                       | TE          | P                     | W            | V                                 |
| Ensley et al. (2006; Sample 2) | 154                           | T, N                               | .20                       | OP                       | TE          | P                     | W            | V                                 |
| Erkutlu (2012)                 | 105                           | A                                  | .39                       | B                        | TE          | P                     | W            |                                   |
| Gupta et al. (2010)            | 28                            | N                                  | .20                       | B, OP                    | TE          | P                     | S            |                                   |
| Gupta et al. (2011)            | 36                            | N                                  | .32                       | B, OP                    | TE          | P                     | S            |                                   |
| Hannah et al. (2011)           | 47                            | N                                  | .39                       | B, SP                    | IN          | P                     | W            | V                                 |
| Hiller et al. (2006)           | 45                            | T, N                               | .29                       | B, SP                    | TE          | P                     | W            |                                   |
| Hmielecki et al. (2012)        | 179                           | N                                  | .37                       | A, B, OP                 | TE          | P                     | W            |                                   |
| Hoch & Kozlowski (in press)    | 101                           | T, N                               | .11                       | SP                       | TE          | P                     | W            | V                                 |
| Hoch (2013)                    | 43                            | N                                  | .37                       | SP                       | TE          | P                     | W            | V                                 |
| Hoch et al. (2010)             | 26                            | A                                  | .45                       | B, SP                    | TE          | P                     | W            |                                   |
| Ishikawa (2012)                | 119                           | C                                  | .42                       | OP                       | PE          | P                     | W            | V                                 |
| Kukenberger et al. (2011)      | 65                            | C                                  | .11                       | OP                       | PE          | U                     | S            |                                   |
| McIntyre & Foti (2013)         | 39                            | C                                  | .31                       | B, OP                    | PE          | P                     | S            |                                   |
| Mehra et al. (2006)            | 28                            | C                                  | .27                       | A, OP                    | PE          | P                     | W            |                                   |
| Mendez (2010)                  | 26                            | T, N                               | .27                       | A, OP                    | PE          | U                     | W            |                                   |
| Moshier & Foti (2012)          | 36                            | C                                  | .19                       | A, B, OP                 | PE          | U                     | S            |                                   |
| Pearce & Ensley (2004)         | 69                            | N                                  | .42                       | A, B, SP                 | IN          | P                     | W            |                                   |
| Pearce & Sims (2002)           | 71                            | T, N                               | .18                       | SP                       | TE          | P                     | W            | V                                 |
| Pearce et al. (2004)           | 28                            | T, N                               | .37                       | B, SP                    | TE          | P                     | W            | V                                 |
| Resick et al. (2012)           | 68                            | T                                  | .15                       | B, OP                    | TE          | U                     | S            |                                   |
| Sivasubramaniam et al. (2002)  | 41                            | T, N                               | .29                       | B, OP                    | TE          | P                     | S            |                                   |
| Small (2007)                   | 59                            | A                                  | .44                       | A, OP, SP                | PE          | U                     | S            |                                   |
| Thompson (2008)                | 8                             | A                                  | .70                       | OP                       | TE          | U                     | W            |                                   |
| Wang et al. (2012)             | 214                           | C, N                               | .39                       | A, B, SP                 | PE, IN      | U                     | W            |                                   |
| Wellman (2013; Sample 1)       | 80                            | C                                  | -.03                      | SP                       | PE          | U                     | W            |                                   |
| Wellman (2013; Sample 2)       | 60                            | C                                  | .24                       | OP                       | PE          | U                     | S            |                                   |
| Zhang et al. (2012)            | 74                            | C, N                               | .15                       | B, OP, SP                | PE, IN      | P                     | W            | V                                 |
| Ziegert (2005)                 | 39                            | C, T                               | .11                       | A, B, OP, SP             | TE          | U                     | W            | V                                 |
| Ziegert et al. (2013)          | 26                            | N                                  | .51                       | A, B                     | TE          | U                     | W            |                                   |
| Ziegert et al. (2008)          | 195                           | N                                  | .25                       | B, SP                    | TE          | U                     | W            | V                                 |
| Ziegert et al. (2009)          | 285                           | N                                  | .16                       | B, SP                    | TE          | U                     | W            | V                                 |
| Zhou (2013)                    | 144                           | C                                  | .45                       | SP                       | PE          | U                     | W            |                                   |

*Note.* For coding the content of shared leadership, T = shared traditional leadership; N = shared new-genre leadership; C = cumulative, overall shared leadership; A = average of various types of leadership (e.g., Hoch et al., 2010, averaged aversive, directive, transactional, transformational, and empowering leadership into one measure of shared leadership). For coding team effectiveness measure, A = attitudinal outcomes; B = behavioral processes and emergent states; OP = objective performance; SP = subjective performance. For coding measurement referent, TE = team as a whole; PE = each peer as the referent; IN = individual members rating themselves. For publication status, P = published in a peer-reviewed journal or a book chapter, U = unpublished studies including dissertation/thesis, conference papers, and working papers. For coding team types, S = student teams; W = work teams. For including vertical leadership, V = includes vertical leadership; blank = does not include vertical leadership.

(Appendices continue)



## Appendix B

### References of the Studies Considered but Ultimately Excluded

#### 1. Education Studies That Confounded Leadership From Multiple Hierarchies in Organizations (Six)

- Davis, M. W. (2009). *Distributed leadership and school performance* (Unpublished doctoral dissertation). The George Washington University, Washington, DC.
- Heck, R. H., & Hallinger, P. (2010). Testing a longitudinal model of distributed leadership effects on school improvement. *Leadership Quarterly*, 21, 867–885.
- Hulpia, H., Devos, G., & Van Keer, H. (2011). The relation between school leadership from a distributed perspective and teachers' organizational commitment examining the source of the leadership function. *Educational Administration Quarterly*, 47, 728–771.
- Leithwood, K., & Jantzi, D. (2000). The effects of transformational leadership on organizational conditions and student engagement with school. *Journal of Educational Administration*, 38, 112–129.
- Leithwood, K., & Mascal, B. (2008). Collective leadership effects on student achievement. *Educational Administration Quarterly*, 44, 529–561.
- Terrell, H. P. (2010). *The relationship of the dimensions of distributed leadership in elementary schools of urban districts and student achievement* (Unpublished doctoral dissertation). The George Washington University, Washington, DC.

#### 2. Individual-Level Studies of Shared Leadership (Four)

- Alves, J. C. (2008). *A multilevel analysis of the association among individual capabilities, team leadership behaviors, and performance in China* (Unpublished doctoral dissertation). University of Illinois at Urbana-Champaign, Champaign, IL.
- Avolio, B. J., Jung, D., Murry, W., & Sivasubramaniam, N. (1996). Building highly developed teams: Focusing on shared leadership process, efficacy, trust, and performance. In M. M. Beyerlein, D. A. Johnson, & S. T. Beyerlein (Eds.), *Advances in interdisciplinary studies of work teams* (pp. 173–209). Greenwich, CT: JAI Press.
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- Davis, C. J. (1995). *Communicating shared vision in a scientific research organization* (Unpublished doctoral dissertation). University of Illinois at Urbana-Champaign, Champaign, IL.

#### 3. Experimental Manipulation on Shared Leadership (One)

- Belau, L., Bell, S. T., & Briggs, A. L. (2008). *Effects of different types of leadership on team performance*. Paper presented at the annual conference of the Society for Industrial Organizational Psychology, San Francisco, CA.

#### 4. Nonindependent Samples (Two)

- Small, E. E., & Rentsch, J. R. (2010). Shared leadership in teams: A matter of distribution. *Journal of Personnel Psychology*, 9, 203–211.
- Ziegert, J. C. (2007). *Does more than one cook spoil the broth? An examination of shared team leadership*. Paper presented at the annual conference of the Academy of Management, Philadelphia, PA.

#### 5. Alternative Conceptualizations of Shared Leadership (Five)

- Acar, F. P. (2010). Analyzing the effects of diversity perceptions and shared leadership on emotional conflict: A dynamic approach. *The International Journal of Human Resource Management*, 21, 1733–1753.
- Li, Y., Chun, H., Ashkanasy, N. M., & Ahlstrom, D. (2012). A multi-level study of emergent group leadership: Effects of emotional stability and group conflict. *Asia Pacific Journal of Management*, 29, 1–16.
- Menges, J. I., Walter, F., Vogel, B., & Bruch, H. (2011). Transformational leadership climate: Performance linkages, mechanisms, and boundary conditions at the organizational level. *Leadership Quarterly*, 22, 893–909.
- Muethel, M., Gehrlein, S., & Hoegl, M. (2012). Socio-demographic factors and shared leadership behaviors in dispersed teams: Implications for human resource management. *Human Resource Management*, 51, 525–548.
- Wood, M. S., & Fields, D. (2007). Exploring the impact of shared leadership on management team member job outcomes. *Baltic Journal of Management*, 2, 251–272.

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